Paing Hein Htet

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EDUCATION

University College London, MEng Biomedical Engineering *Sept 2025 – Jun 2029*

First-year result: First Class. Second year in progress. **Memberships:** IEEE (Student Member), IET (Student).
A-Levels (2025): A* Maths, Further Maths, Physics, Chemistry; A Biology. **IGCSE** (2023): grade 9 in all nine subjects.

PUBLICATIONS & PREPRINTS

Duplicate leakage in a public conjunctival pallor benchmark, and an honest baseline for image-based anaemia screening

Preprint. Htet, P. H. (2026). Zenodo. doi.org/10.5281/zenodo.22782147. Sole author. Code: github.com/Pai-88/pallor-hb

- Showed 710 records in a published clinical image dataset are really 383 photographs; the copies cross sites with conflicting labels, so site-grouped folds, the standard safeguard, still left two-fifths of a **+0.176** AUROC inflation.
- Corrected baseline: AUROC **0.706** (95% CI 0.653–0.757); every number regenerated by script and verified automatically (68 claims); code and detector released.

PROJECTS

PURSUIT-1: Person-Following Quadcopter *2026*

- Designed the 4-layer board that is also the airframe, from a Python netlist; rev A fabricated and assembled.
- Wrote the ESP32-S3 flight firmware: sensor fusion, cascaded PID, motor mixing and failsafes. Bench-tested, not yet flown.
- Camera pipeline for detection, tracking and face recognition: **0 false accepts in 24,110 impostor faces**; 369 tests.

KiCad and its Python API, ngspice, Python, C++, ONNX Runtime, OpenCV, PlatformIO. Code: github.com/Pai-88/pursuit-drone

EIT-16CH: Electrical Impedance Tomography Instrument *2026*

- Designed a 16-electrode 4-layer measurement board, 100 components, DRC clean; not yet fabricated.
- Compared learned and classical image reconstruction in simulation: a linear learned inverse wins only below 40 dB, a U-Net at every noise level tested (**3.3 times** the classical score at 20 dB).

KiCad, Freerouting, ESP32-S3, PlatformIO, PyTorch, NumPy, pyEIT.

TT-Scout: Table Tennis Match Analysis from Phone Video *2026*

- Finds every point and its winner in one phone video and measures 3D shot speed, calibrated from the table itself.
- Point winners correct on all **21** endings judged by eye, 5 of them blind; median speed error 3% on simulated shots.

Python, OpenCV, NumPy, PyTorch. Code: github.com/Pai-88/tt-scout

RL Car: Uncertainty-Aware Autonomous Racing *2026*

- Showed, in simulation, that uncertainty must be estimated on a forward model, not the policy: the policy-space ensemble scored **AUROC 0.127**, anti-correlated, on a blinded sensor.
- Isolating the faulty sensor instead of braking cut crashes from 155 to **2** in 192 held-out fault episodes at full speed.
- Wrote a Navier–Stokes solver from scratch for the body aerodynamics, validated on published cylinder-flow benchmarks.

Python, NumPy, SciPy, bootstrap ridge regression, random Fourier features; FreeCAD, gmsh, CalculiX.

Hand-Crank Phone Charger: Printed Planetary Drive, Stepper Motor as Alternator *2026*

- Designed and printed a 5.35:1 herringbone planetary stage, printed in place, that drives a NEMA23 stepper as an alternator.
- SPICE model predicts **10 W** at the phone for 17.5 W at the hand at 90 rpm, written down before the first test on 25 Oct 2026.

Fusion 360, Blender, ngspice, Python; PETG on a Bambu P1S. Design document at paingheinhtet.com

RESEARCH

Surgical video segmentation, with Dr Sophia Bano, UCL Computer Science *from Oct 2026*

Oral cancer detection with deep learning, with Dr J. W. Asare, UENR, Ghana *2026 – present*

WORK EXPERIENCE

Healthcare Assistant (internship), Grand Mandalay Hospital, Mandalay, Myanmar *Jan 2025 – Jun 2025*

- Six months on the wards, supporting patient care and observations; it turned me towards clinical machine learning.

TECHNICAL SKILLS

Languages and CS: Python, C, C++, MATLAB, SQL; data structures and algorithms.

Statistics: bootstrap CIs, DeLong test, Hosmer–Lemeshow calibration, expected calibration error, Brier score, Bland–Altman agreement, decision curves, grouped cross-validation.

Machine learning: epistemic uncertainty via bootstrap ensembles and random Fourier features, out-of-distribution detection, model calibration, evaluation design and leakage control, reinforcement learning (MuJoCo, domain randomisation), gradient boosting; scikit-learn, PyTorch.

Deep learning and vision: MLP, CNN, RNN/LSTM, Transformers, ViT, autoencoders; DETR, YOLO, OpenCV, ONNX Runtime, CoreML; detection, tracking, face recognition.

LLMs and agents: Anthropic API, tool use, MCP, multi-step agents, Hugging Face Transformers, Whisper, smolagents.

Data and signals: pandas, NumPy, SciPy, Matplotlib, Tableau; dataset-integrity auditing; EMG, PPG and Wi-Fi channel-state processing.

Electronics: 2- and 4-layer PCB layout, analog front ends, power; KiCad (Python API), LTspice, ngspice; soldering; ESP32-S3, RP2040.

CAD and laboratory: Fusion 360, SolidWorks, CadQuery, 3D printing, Git, LaTeX; aseptic technique, cell passaging, Instron tensile testing.